

Weapon Detection CCTV System — YOLO11 Pipeline

Stages 03–09: Sample Generation, Training Strategy, Baseline, HPO Framework, Final Training, Error Analysis & Improved Fine-Tuning

Project (Weapon130): Real-world intelligent CCTV surveillance system for weapon detection, with a future multi-camera tracking / re-identification integration target.

Base architecture: YOLO11 (Ultralytics), transfer-learned from COCO-pretrained weights — never trained from scratch, at any stage.

Dataset used throughout this phase: sample_dataset — a 20,000-image representative sample ($\approx 18.3\%$) of the 109,403-image cleaned corpus, validated at 99.59% overall representativeness.

Scope of this report: notebooks 03 through 09 of the pipeline — from representative sample generation through the final gate decision to proceed to full-dataset (130K) training.

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1. Executive Summary

This report consolidates seven engineering notebooks (03–09) covering the middle phase of the Weapon130 CCTV weapon-detection project: building a statistically representative training sample, designing an evidence-based training strategy, running a baseline fine-tune, building (but not yet executing) a hyperparameter-optimization framework, running the official 'final' training on the sample, diagnosing why that run regressed, and implementing a fix that ultimately cleared the project's gate for full-dataset training.

Headline result: the corrected model (EXP-IMPROVED-09) beats the original baseline (EXP-01) on every core metric — precision +10.2 points, recall +12.3 points, mAP50 +15.3 points, mAP50-95 +13.4 points — and clears the validation gate (mAP50-95 ≥ 0.315) with a 42.6% margin (0.4491 vs. 0.315), unlocking Stage 10 (full 130,000-image training).

Journey at a glance

Stage	What happened	Precision	Recall	mAP50	mAP50-95	Verdict
03	Built 20,000-image representative sample (99.59% score)	—	—	—	—	Approved for HPO/tuning
04	Defined training strategy (imgsz=800, YOLO11s, AdamW, class-weighted loss)	—	—	—	—	Reference config only, no training
05 (EXP-01)	Baseline fine-tune, 100 epochs	0.633	0.485	0.503	0.315	Baseline established
06	HPO framework built; 10 planned experiments NOT executed	0.633	0.485	0.503	0.315	EXP-01 remains default (only completed run)
07 (EXP-FINAL-07)	'Official' final run, same config as EXP-01	0.633	0.456	0.482	0.297	REGRESSED — Stage 08 gated
08	Root-cause diagnosis of the regression	—	—	—	—	8-item improvement roadmap issued
09 (EXP-IMPROVED-09)	Unfreeze backbone + imgsz 960 + wider mosaic tail	0.735	0.608	0.656	0.449	GATE PASSED — ready for Stage 10

Two structural facts run through the whole report: (1) every fine-tuning run in this phase uses the 20,000-image representative sample, not the full 109,403-image cleaned dataset — the full-dataset generalization gate (EXP-10 in the Stage 04 roadmap) is still outstanding and independent of the gate this report tracks; and (2) the Stage 06 hyperparameter-optimization framework was built and validated as code but its ten planned ablations (EXP-02...EXP-11) were never actually run on GPU — this gap is exactly what caused Stage 07 to blindly re-ship the untested baseline configuration, which then underperformed and triggered the Stage 08/09 recovery work.

2. Notebook 03 — Representative Sample Generation

Input: clean_dataset/ (109,403 images, 294,950 boxes across train/valid/test — the output of Stage 02 cleaning). Goal: build a small, statistically representative subset so that architecture search, augmentation tuning, and hyperparameter sweeps can be iterated on in minutes instead of hours, without biasing results relative to the full dataset. The source dataset is never modified — every operation is read-only on the source and copy-based on the output.

2.1 Sample Size Decision

Five candidate sizes (5K/10K/20K/30K/50K) were compared on three axes: statistical representation (using the rarest stratum, Blunt_Weapon at ~3.2% of boxes, as the binding constraint), expected training time on a single RTX-4090-class GPU, and expected HPO ranking quality.

Decision: TARGET_SAMPLE_SIZE = 20,000 images. 10K and below failed the instance-floor test for rare classes; 30K/50K improved representativeness only marginally while roughly doubling/quadrupling training cost.

2.2 Sampling Method

Not random sampling. The algorithm is deterministic proportional stratified sampling with secondary-axis-aware tie-breaking:

- Primary stratification key: split × dominant_class × size_bucket, with largest-remainder rounding so per-stratum allocations sum exactly to 20,000.
- Within each stratum, candidate images are ranked by a diversity score (under-representation of camera_proxy) and the highest-diversity images are taken first — a deterministic farthest-first-style selection.
- Shortfalls in very rare corners (e.g. test × Blunt_Weapon × small) are redistributed proportionally across the remaining strata of the same class.
- RANDOM_SEED = 42 is used only for stable tie-breaking, never for primary selection — re-running is fully reproducible.

Constraints explicitly preserved by construction: class ratios, bounding-box ratios, object-size distribution, image resolution, train/val/test ratios, camera/source diversity, indoor/outdoor & lighting diversity, occlusion (multi-object) diversity, blur diversity, and background (empty-label) ratio.

2.3 Representativeness Score & Recommendation

Each comparison axis is scored as 1 – normalized L1 distance between full-dataset and sample distributions (100% = identical). The count-weighted overall score across the four core axes (class balance, object size, box/split ratio, image distribution) confirmed the sample met its design target (formally reported as 99.59% in Notebook 04, Section 1).

Engineering recommendation: Yes, use for HPO (only relative ranking between configurations is needed). Yes, use for the first fine-tuning experiments for pipeline/recipe validation — but not as the reported production numbers; retrain the winning configuration on the full clean_dataset before deployment. Confidence in the sample's conclusions is high for relative rankings, moderate for absolute metric values — expect the sample-trained model to read 2–5 mAP points higher than a full-dataset model, and rare-class AP gaps of 5–10 points, purely as sample-size effects.

3. Notebook 04 — Dataset Optimization & Training Strategy

This notebook performs no model training. Its sole purpose is to produce a reproducible, evidence-based engineering configuration — image size, augmentation policy, class-imbalance strategy, architecture choice, hyperparameters, and a 10-experiment roadmap — governing the Stage 05 baseline fine-tune.

3.1 Sample Validation (Section 1)

Overall weighted representation score: 99.59%. Class balance (99.21%) and object-size distribution (99.61%) — the two axes that matter most for detector training — are essentially preserved. Split ratio and resolution match 100.0% (uniform 800×800 post-cleaning). The one weak axis is Camera/Source Coverage at 70.05% — flagged as a caveat requiring re-validation once the model nears deployment quality, but not disqualifying for the first experiments.

3.2 Image Size (Section 2)

Native resolution: 800×800. Average bounding box: 273×282 px (~34–35% of frame edge); only 1.33–1.37% of boxes are 'small', 84.6–85.0% are 'large' — a large-object-dominant dataset, unlike COCO.

Recommendation: imgsz = 800. 640 buys ~22% faster training for an estimated 3–4 point mAP cost concentrated on small/medium objects — a bad trade for a weapon detector where small/medium instances are the safety-critical cases. 960 gives only +0.5–1.5% mAP for ~35% slower training and near-total VRAM consumption (~14GB of 16GB budget) — kept only as a later ablation.

3.3 Small-Object Risk (Section 3)

Small objects are 1.37% of boxes by rate but ~4,040 boxes in absolute count across the full corpus — rare in rate, not negligible in count. Small-object detection is judged a secondary risk; the dominant challenges are class imbalance and rare-class/small-object co-occurrence.

Recommendation: single-scale training at imgsz=800; multi-scale training retained only as a targeted ablation (Experiment 6) if empirical recall gaps appear later.

3.4 Augmentation & Class-Imbalance Strategy (Sections 4–5)

Dataset profile driving every augmentation choice: large-object-dominant, 12.4× class imbalance, fixed 800×800 imagery, 61–63% multi-object images, 73–74% normal lighting / 11–15% low-light / 12–15% bright, 99.97% sharp.

Decision: class-weighted loss (~2–3× on Blunt_Weapon, Person, Tool) combined with rare-class-biased augmentation (Copy-Paste=0.1, moderate HSV). Weighted loss was chosen over Focal Loss for its directness and zero extra cost; manual oversampling was explicitly rejected due to a mosaic-duplication risk flagged in the Stage 02 report. Additional data collection was endorsed as a parallel, non-blocking initiative.

3.5 Architecture (Section 6)

Hardware assumption: single NVIDIA GPU, 16GB VRAM.

Recommendation: YOLO11s. YOLO11n under-fits given the rare-class recall requirement; YOLO11m/l/x fit in 16GB but their accuracy gains are likely modest here because the dominant challenge is class imbalance, not model capacity. YOLO11s leaves batch-size ≥16 headroom at imgsz=800 and remains real-time-viable for the eventual multi-camera CCTV target; YOLO11m is reserved as a later accuracy-ceiling ablation.

3.6 Fine-Tuning Configuration (Section 7)

Parameter	Value	Rationale (summary)
Optimizer	AdamW	Faster/more stable convergence on transfer-learning fine-tunes; Ultralytics default
Learning rate	0.001 → cosine to 1%	Conservative LR for fine-tuning a pretrained backbone
Weight decay	0.0005	Ultralytics default; augmentation already regularizes
Momentum	0.937	Standard, well-validated YOLO default

Scheduler	Cosine (cos_lr=True)	Smooth decay for the 150-epoch budget
Warmup	3 epochs	Stabilizes frozen-backbone unfreeze + mosaic transition
Epochs	150 (patience=30)	Balances convergence against compute budget
Batch size	16	~40% VRAM headroom on YOLO11s @ imgsz=800
Workers	8	Matches typical desktop CPU cores; Windows dataloader overhead
Cache	ram	20,000×800×800×3 images fit in typical 32–64GB RAM
AMP	True	Halves VRAM, negligible accuracy cost
Freeze	10 backbone layers	Preserve COCO features while head adapts to 7 classes
EMA	Enabled	Stabilizes final weights against batch noise
Rect training	False	Dataset is 100% square 800×800; no benefit
Close-Mosaic epoch	Last 10 epochs	Fine-tune on undistorted images near convergence
Seed	42	Matches Stage 03 seed for end-to-end reproducibility

3.7 Experiment Roadmap (Section 8) & Cost Estimate (Section 9)

Ten single-variable experiments were planned, ordered by priority: EXP-01 (baseline) → EXP-02 (backbone freezing) → EXP-03 (class weighting) → optimizer, batch size, resolution/scale, augmentation policy, architecture ceiling → EXP-10 (full-dataset generalization gate) before any deployment candidacy. All Section 9 cost figures (training time, VRAM) are explicitly stated as engineering projections, not measurements — no training had been run at this point in the pipeline.

4. Notebook 05 — Experiment Management & Baseline Fine-Tuning (EXP-01)

This notebook is the permanent experiment-management framework for every YOLO experiment in the project, not a one-off script. Stage 4's JSON files (`recommended_hyperparameters.json`, `training_strategy.json`) are the single source of truth for every hyperparameter — the only hand-set fields are per-run identity (`experiment_id`, `name`, `description`) and machine-local infra (`device`, `project_dir`).

4.1 Framework Components

- Section 1: loads and cross-validates the two Stage 04 config files, builds the `ExperimentConfig` used everywhere downstream.
- Section 2–3: auto-generates an isolated `experiments/EXP-01/` directory tree and writes full reproducibility artifacts (config, git state, environment, package list, hardware info) before any training step.
- Section 4: validates `sample_dataset` end-to-end (YAML resolution, paired image/label counts, valid class IDs, no zero-byte files) — training only proceeds if this passes.
- Section 5: loads COCO-pretrained `yolo11s.pt` (fine-tune only, never random init), freezes the first 10 backbone layers, and auto-resumes from `last.pt` if a previous run was interrupted.
- Section 6: runs the baseline fine-tune with exactly the Stage 04 hyperparameters — no search, no modification.

4.2 EXP-01 Baseline Results

Metric	Value
Precision	0.6327
Recall	0.4847
mAP50	0.5026
mAP50-95	0.3150
Training time	11.37 hours
GPU memory	3.29 GB
Model / Optimizer	yolo11s.pt / AdamW
Epochs / imgsz / batch	100 / 800 / 16

Per-class results (val split)

Class	Precision	Recall	mAP50	mAP50-95
Blunt_Weapon	0.507	0.331	0.324	0.216
Explosive	0.690	0.557	0.583	0.367
Fire_Smoke	0.658	0.550	0.569	0.303
Firearm	0.742	0.687	0.713	0.450
Melee_Weapon	0.658	0.651	0.662	0.437
Person	0.539	0.307	0.295	0.175
Tool	0.635	0.308	0.374	0.258

4.3 Automatic Engineering Recommendation (Section 12)

- Baseline healthy: MARGINAL/NO — mAP50-95 below the expected range; investigate data quality, labels, or epoch count.
- Underfitting: POSSIBLE — training loss comparatively high; consider more epochs or reduced freeze.
- Overfitting: YES — meaningful train/val loss divergence in the final epochs.
- Begin HPO: NOT YET — stabilize the baseline first.
- Additional data cleaning needed: NO — Stage 1–3 cleaning and 99.59% representativeness are sufficient for this tier.

Priority for next work: recall is materially lower than precision — prioritize a class-weighted-loss ablation, since missed weapon detections are the highest-cost failure mode for a surveillance system.

5. Notebook 06 — Hyperparameter Optimization Framework

This notebook is explicitly not a random hyperparameter search — it is a controlled, single-variable experiment-management framework: diagnose the baseline's weaknesses, design ~10 experiments changing exactly one hyperparameter each, run them all through a reusable ExperimentManager, rank them automatically, and recommend one configuration for final training.

5.1 Baseline Diagnosis (Section 1)

Confirms Blunt_Weapon, Person, and Tool — despite already receiving the largest class weight (2.5×) — remain the weakest classes, pointing at augmentation/regularization/optimizer tuning rather than further loss re-weighting.

5.2 Planned Roadmap (Section 2): EXP-02...EXP-09

Exp.	Parameter	From → To	Priority
EXP-02	lr0	0.001 → 0.0005	Critical
EXP-03	lr0	0.001 → 0.002	High
EXP-04	batch	16 → 24	Medium
EXP-05	weight_decay	0.0005 → 0.001	Medium
EXP-06	optimizer	AdamW → SGD	High
EXP-07	freeze	10 → 0	Critical
EXP-08	imgsz	800 → 960	Medium
EXP-09	mosaic	1.0 → 0.5	Medium

5.3 Execution Status — Critical Gap

None of the ten planned experiments were executed. The notebook's master switch EXECUTE_TRAINING defaulted to False, so the ExperimentManager, ranking engine, and visualizations were built and validated as code, but only ever ran against the single pre-existing EXP-01 result already on disk.

Section 6's automatic ranking therefore had exactly one row to rank: EXP-01 was declared 'not Pareto-dominated' and selected as the Final Training configuration by default — not because it beat any real alternative, but because no alternative was ever tried. This gap becomes the root cause investigated in Notebook 08.

6. Notebook 07 — Optimized (Final) Fine-Tuning on the Sample

Trains the 'official final' YOLO11 model on sample_dataset using the Stage 06-selected configuration (EXP-01, unchanged, since it was never actually challenged). No further hyperparameter search is performed — Stage 06 is treated as the single source of truth.

6.1 Result: EXP-FINAL-07

Metric	Baseline (EXP-01)	Final (EXP-FINAL-07)	Delta	% Change
Precision	0.6327	0.6333	+0.0006	+0.09%
Recall	0.4847	0.4557	-0.0290	-5.99%
mAP50	0.5026	0.4815	-0.0211	-4.20%
mAP50-95	0.3150	0.2973	-0.0177	-5.61%
Training time (h)	11.373	11.313	-0.060	-0.53%

Weakest classes (unchanged): Tool, Person, Blunt_Weapon (by recall)

6.2 Verdict

Deployment readiness: NOT YET READY. mAP50-95 regressed materially against the baseline; the notebook explicitly recommends holding Stage 08 (full 130K training) pending investigation, and flags that the Stage 04 full-dataset generalization gate (EXP-10 on clean_dataset) remains outstanding regardless of this run's verdict.

7. Notebook 08 — Error Analysis & Model Improvement Strategy

A pure diagnosis notebook: no training, no dataset modification, no re-run of HPO. Its only output is a prioritized, evidence-linked improvement roadmap, built entirely by re-reading Stage 07's own artifacts (metrics, training curves, confusion matrices, PR/F1 curves) against the Stage 06 roadmap.

7.1 Aggregate Comparison

Precision improved marginally (+0.09%, essentially a reproduction). Recall regressed ~6.0%, mAP50 ~4.2%, mAP50-95 ~5.6% — three of four core metrics got worse using the supposedly 'best' configuration. Training time was flat, ruling out a shortened run as the cause.

7.2 Per-Class Findings

- Strongest: Firearm (mAP50 0.712) and Melee_Weapon (0.662) — high-count, visually distinctive classes.
- Weakest: Person (0.295), Blunt_Weapon (0.323), Tool (0.373) — exactly the three classes carrying the highest class weight (2.5×), a strong signal that class weighting alone is not fixing a root data/representation problem.
- Person has unusually low support (~100 predicted instances vs. 1,746 for Firearm) — points to either a genuinely rare class or an annotation/labeling gap.

7.3 Confusion Matrix Findings

The largest confusions form a triangle: Melee_Weapon ↔ Firearm (27/25), Melee_Weapon ↔ Tool (22), Melee_Weapon ↔ Blunt_Weapon (19). This is diagnosed as a fine-grained, semantically adjacent taxonomy problem (a hammer, crowbar, or machete could reasonably be filed under any of three labels) — possibly compounded by inconsistent annotation guidelines, not a model-capacity problem that more training alone would fix.

7.4 Training Curve Findings

- Convergence: stable, smooth, no divergence over 100 epochs.
- Overfitting: not evident — val loss stays above train loss throughout with no upward inflection.
- Plateau: recall and mAP flatten in the final 30–40% of training; a cls-loss step-down near epoch 90 (coincident with close_mosaic) is not matched by validation gains — the model reached a data/config ceiling well before epoch 100.

7.5 Root-Cause Conclusion

None of the ten Stage 06 experiments were ever executed before EXP-01 was declared the winning configuration. Stage 07 therefore trained on a config that was never empirically compared to its own hypothesized alternatives. Ranked causal candidates: (1) freeze=10 — consistent with 'headroom, not overfitting'; (2) imgsz=800 — consistent with Person/Tool being weakest; (3) lr0=0.001 unbracketed — consistent with the late-training plateau; (4) the mosaic/close_mosaic schedule — consistent with the plateau timing.

7.6 Improvement Roadmap (8 items, priority order)

#	Action	Target
1	Unfreeze the backbone (freeze=0)	Recall, mAP50-95
2	Increase resolution 800 → 960 (reduce batch to compensate)	Person, Tool recall
3	Bracket the learning rate (0.0005 and 0.002 vs. 0.001)	mAP50-95, plateau removal
4	Audit annotation guidelines for Blunt_Weapon/Melee_Weapon/Tool	Confusion-triangle precision
5	Per-class confidence thresholds (zero retraining cost)	Deployment-time P/R trade-off
6	Targeted data collection for Person specifically	Person recall & mAP50
7	Test close_mosaic tail length (10→20) or lighter mosaic (1.0→0.5)	mAP50-95 late-training
8	Actually run EXP-02...EXP-12 on the full sample	Validates items 1–3, 7 with real evidence

Explicitly NOT recommended: more epochs at the same config (a genuine plateau, not an in-progress climb), switching optimizer (no supporting evidence), increasing weight decay (no overfitting signature to correct), or broader resampling (representativeness is already 99.59%; the one sub-99% axis is a separately tracked, unrelated gap).

Caveat carried forward: if mAP50-95 does not clear the 0.315 baseline after implementing items 1–3, that would point to a data-side root cause (items 4/6) rather than a hyperparameter-side one.

8. Notebook 09 — Improved Fine-Tuning & Final Gate Decision

Implements only the approved, evidence-linked Stage 08 changes — no new hyperparameter search, no strategy redesign, no change to anything Stage 08 did not flag. Six fields were changed from the EXP-01/EXP-FINAL-07 configuration: unfreeze (freeze 10→0), resolution (imgsz 800→960, batch 16→12 to compensate), learning-rate bracket testing, and the mosaic/close_mosaic schedule adjustments (roadmap items 1, 2, 3, 7). Everything else — optimizer, weight decay, cosine LR, warmup, momentum, patience, class weights, seed, and the pretrained checkpoint — was carried over unchanged.

8.1 Result: EXP-IMPROVED-09

Metric	Baseline (EXP-01)	Final (EXP-FINAL-07)	Improved (EXP-IMPROVED-09)	Δ vs. Baseline
Precision	0.6327	0.6333	0.7350	+0.1023
Recall	0.4847	0.4557	0.6077	+0.1230
mAP50	0.5026	0.4815	0.6555	+0.1529
mAP50-95	0.3150	0.2973	0.4491	+0.1341

Training time increased from 11.37h to 15.32h — an expected, accepted tradeoff of the imgsz 800→960 change (~1.4× compute/step) combined with the batch 16→12 reduction (~1.3× more steps/epoch), not a regression signal and not part of any Stage 08 gate.

Per-class mAP50 / recall — biggest movers

Class	mAP50 (EXP-01)	mAP50 (Improved)	Δ mAP50	Recall (EXP-01)	Recall (Improved)	Δ Recall
Blunt_Weapon	0.324	0.461	+0.137	0.331	0.451	+0.120
Person	0.295	0.421	+0.126	0.307	0.459	+0.152
Tool	0.374	0.489	+0.115	0.308	0.478	+0.170
Melee_Weapon	0.662	0.701	+0.039	0.651	0.688	+0.037
Fire_Smoke	0.569	0.601	+0.032	0.550	0.592	+0.042
Explosive	0.583	0.612	+0.029	0.557	0.601	+0.044
Firearm	0.713	0.742	+0.029	0.687	0.715	+0.028

The three previously-weakest classes (Blunt_Weapon, Person, Tool) improved the most in absolute terms — directly consistent with the imgsz 800→960 change (roadmap item 2) targeting exactly these classes.

8.2 Final Gate Decision

Gate check	Value
Gate metric	mAP50-95
Gate threshold (EXP-01 baseline)	0.3150
EXP-IMPROVED-09 value	0.4491
Margin	+0.1341 (+42.56%)
Gate passed	TRUE

- Q1 — Did the Stage 08 recommendations improve the model? YES — 4/4 core metrics improved vs. EXP-FINAL-07.
- Q2 — Is it now better than the EXP-01 baseline? YES — 4/4 core metrics improved.
- Q3 — Ready for full-dataset (130K) training? YES — per the explicit Stage 08 gate (mAP50-95 ≥ 0.315).

Overall readiness for Stage 10: READY.

8.3 Open Items Carried Forward (gate passing does not close these)

- Annotation-guideline review for Blunt_Weapon / Melee_Weapon / Tool — the confusion triangle is unchanged.

- Targeted Person data collection — Person remains the weakest or near-weakest class by mAP50 despite the largest relative gain.
- Roadmap item 8 was not done: EXP-02...EXP-12 were never run individually on the full sample, so this run bundled several changes together and individual contribution sizes are not isolated.
- The Stage 04 full-dataset generalization gate (EXP-10 on clean_dataset, 109,403 images) is still required before formal deployment candidacy — this notebook's gate concerns the representative sample only.

9. Consolidated Metrics Timeline

Run	Notebook	Precision	Recall	mAP50	mAP50-95	Training time
EXP-01 (baseline)	05	0.633	0.485	0.503	0.315	11.37h
EXP-FINAL-07	07	0.633	0.456	0.482	0.297	11.31h
EXP-IMPROVED-09	09	0.735	0.608	0.656	0.449	15.32h

Reading the timeline: EXP-01 established the baseline. EXP-FINAL-07 was meant to be a confirming 'official' run of the same, HPO-selected configuration — but because the HPO framework's ten planned ablations were never executed, it silently re-shipped an untested config and regressed on three of four metrics. Notebook 08 diagnosed this analytically (no training) and produced a targeted, evidence-linked fix. EXP-IMPROVED-09 applied that fix and not only recovered the regression but substantially exceeded the original baseline on every metric, clearing the gate for full-dataset training.

10. Overall Conclusions & Recommendations

10.1 What worked well

- Stratified, diversity-aware sampling (Notebook 03) produced a 20,000-image sample validated at 99.59% representativeness — a defensible basis for every downstream decision in this phase.
- Every training decision in Notebook 04 (image size, architecture, loss strategy, hyperparameters) was tied to specific dataset evidence rather than defaults, and clearly separated from the training that hadn't happened yet.
- The experiment-management framework (Notebooks 05/06) enforces reproducibility (isolated folders, config snapshots, environment capture, registry rows) and treats Stage-04 config files as the single source of truth — a strong MLOps discipline.
- The error-analysis process (Notebook 08) is a model of evidence-based debugging: every recommended change is traced to a specific piece of observed evidence (loss curves, confusion matrix, per-class metrics) rather than guesswork.
- The final fix (Notebook 09) worked decisively: all four core metrics improved by 10–15 points over the original baseline, and the improvement was concentrated exactly where predicted (Person, Tool, Blunt_Weapon).

10.2 Key risk / process gap identified

The single most consequential issue in this phase: Notebook 06 built a complete hyperparameter-optimization framework but never executed it (`EXECUTE_TRAINING=False`). Notebook 07 then trusted the ranking output of that framework — which had exactly one data point — and shipped an unvalidated configuration as the 'official final' model. This directly caused the Stage 07 regression and the extra Notebook 08/09 recovery cycle. The lesson for future stages: a framework that can rank experiments is not the same as having run the experiments; 'best configuration by composite score' is only meaningful once more than one configuration has actually been tried.

10.3 Outstanding items before production deployment

- Full-dataset generalization gate (Stage 04's EXP-10): every result in this report is on the 20,000-image sample, not the full 109,403-image `clean_dataset`. This must be re-validated before deployment.
- Annotation-guideline audit for the Blunt_Weapon / Melee_Weapon / Tool confusion triangle — this is a labeling problem, not something further training will resolve.
- Targeted data collection for the Person class, which remains the weakest class despite carrying the maximum class weight and gaining the most from the resolution increase.
- Isolating the individual contribution of each of the four bundled changes in EXP-IMPROVED-09 (unfreeze, resolution, LR, mosaic schedule) by finally running the Stage 06 single-variable experiments — currently the gains are proven collectively but not individually attributed.
- Camera/Source Coverage remains the one representativeness axis below 99% (70.05%) — a known, tracked, not-yet-closed gap.

10.4 Recommended next step

Proceed to Stage 10 (full 130K-image training) using the EXP-IMPROVED-09 configuration as the starting point, since the Stage 08 validation gate has been passed with a substantial margin. In parallel — not blocking Stage 10 — begin the annotation audit and Person-class data collection work, and schedule the individual single-variable re-runs (Stage 06 roadmap item 8) to properly attribute the gains achieved in this phase before any final production sign-off.